

Supplementary Material

Figure S1: Gene variance distribution in the PRAD dataset. The dashed line indicates the selected variance threshold (1.0).

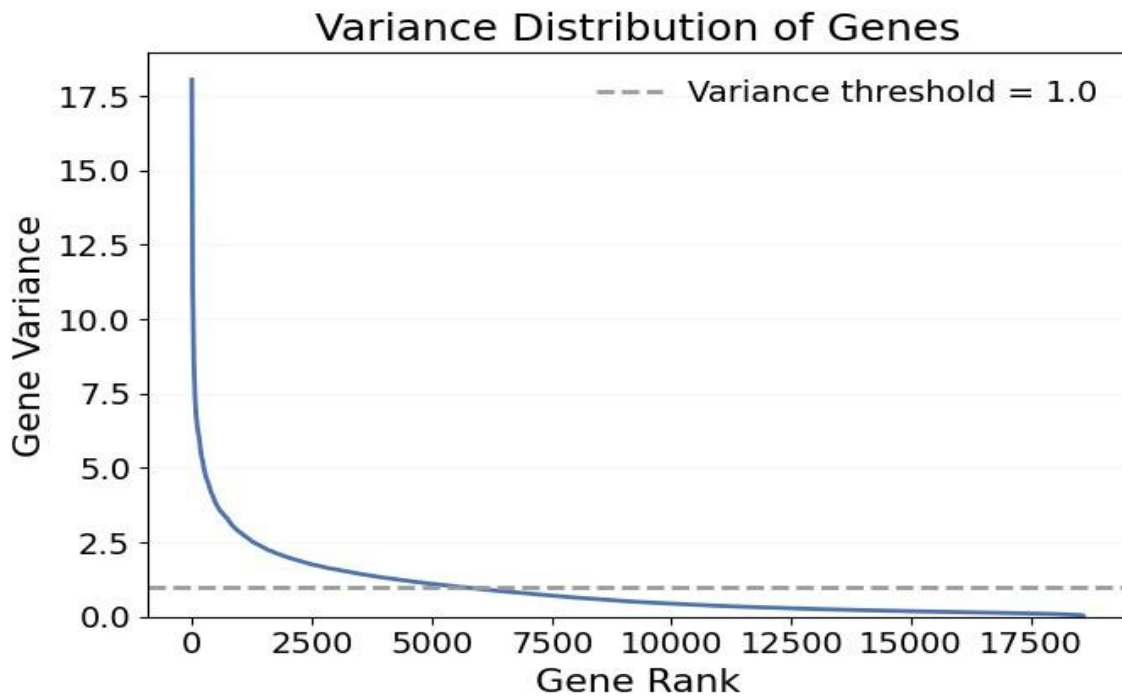


Figure S2: Training and validation loss curves of the autoencoder model.

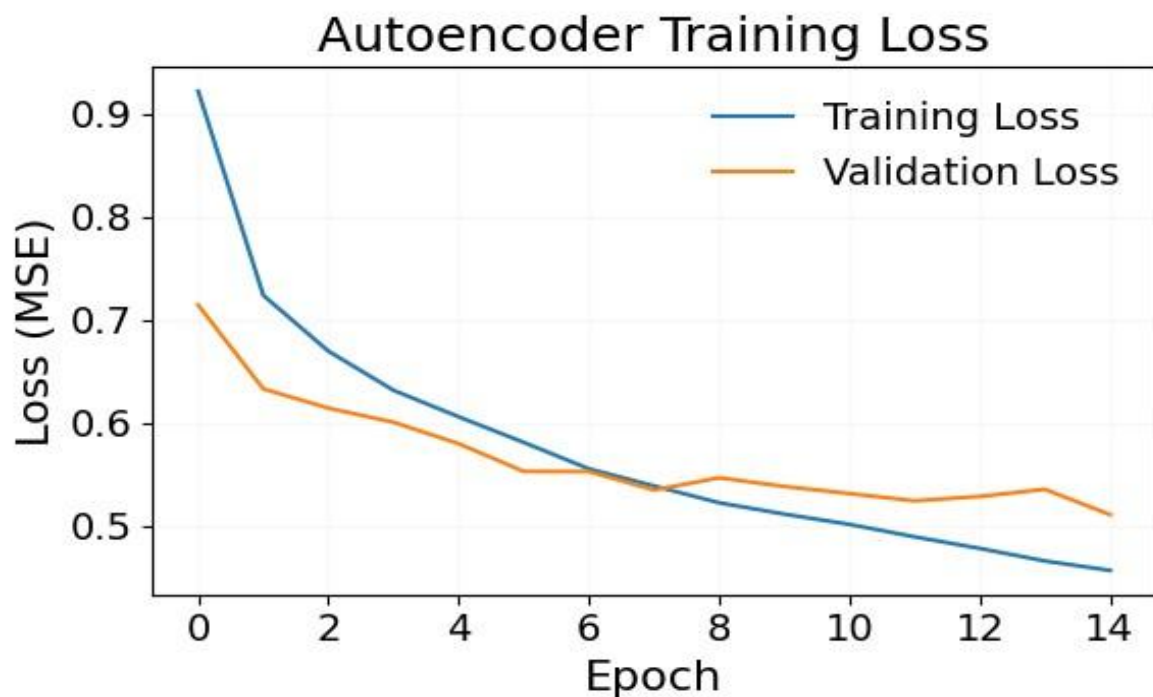


Figure S3: Silhouette analysis indicating $K = 2$ as the optimal number of clusters.

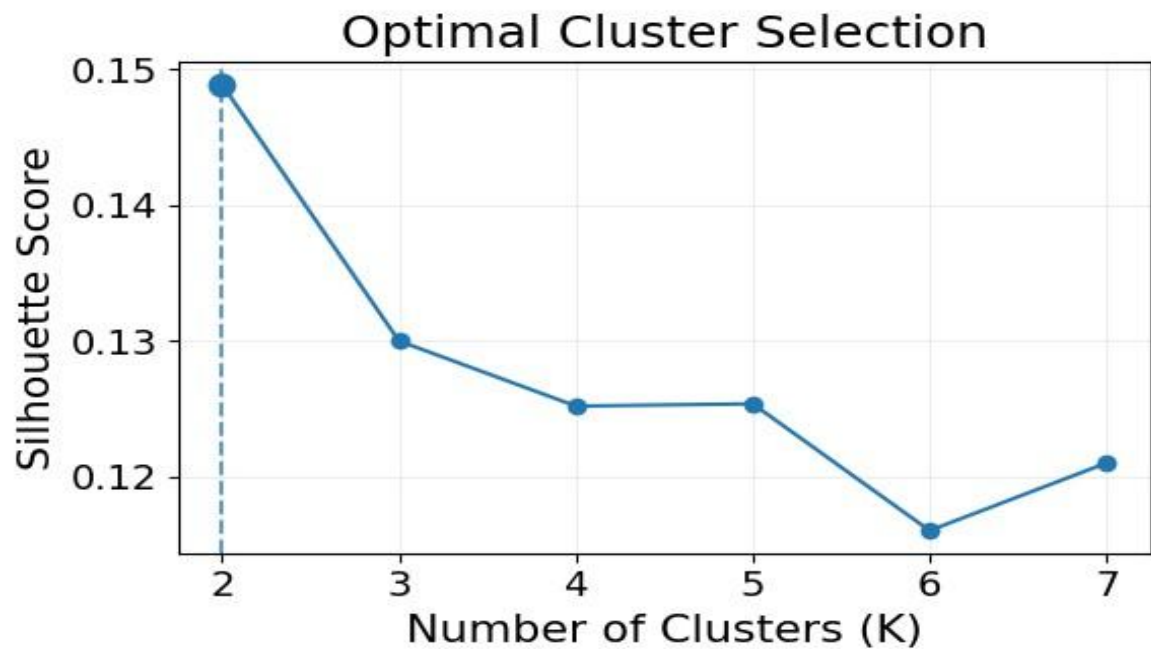


Figure S4: KEGG pathway enrichment analysis of significant differentially expressed genes.

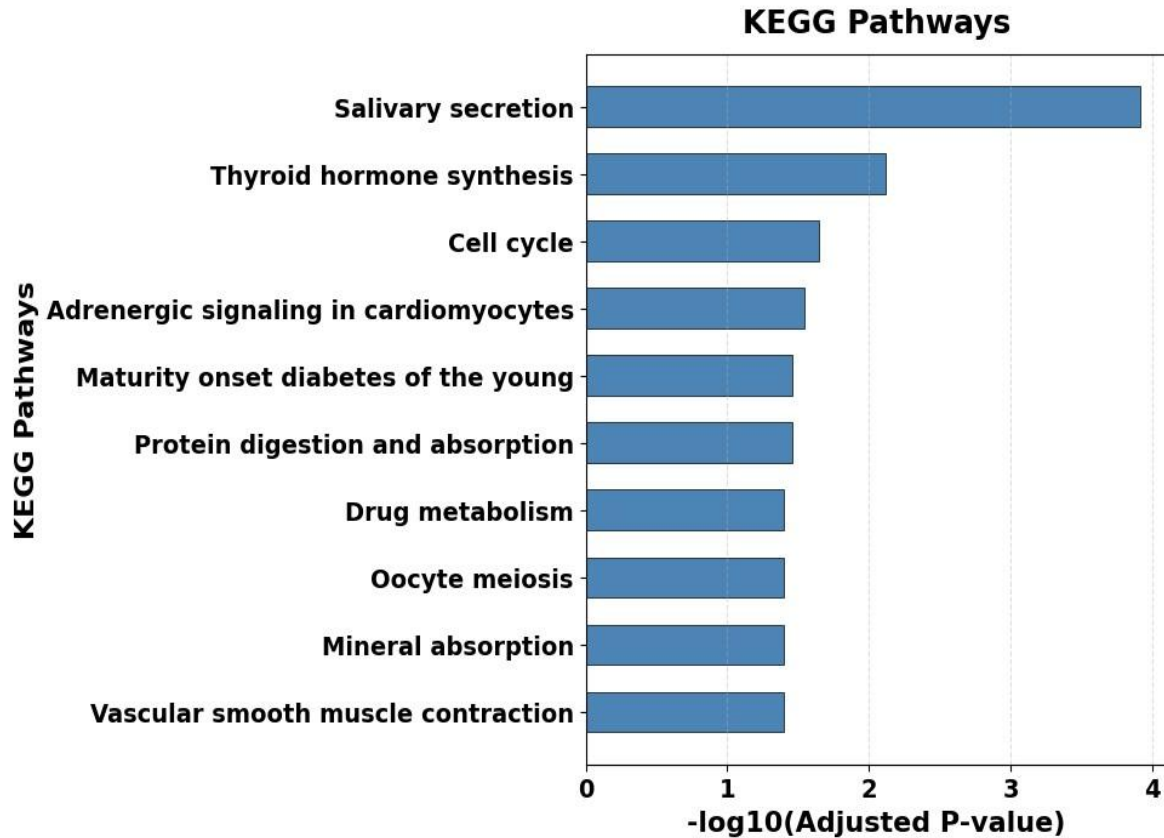


Figure S5: GO biological process enrichment analysis of significant differentially expressed genes.

